Signly: An American Sign Language Learning Application

Abdul Rahim Mohsin, Samad Khan, and Saif Siddiqui

Department of Computer Science, Ontario Tech University

Author Note

Abdul Rahim Mohsin (100890247), Samad Khan (100754092), and Saif Siddiqui (100824868), Department of Computer Science, Ontario Tech University.

This report was completed in fulfillment of the final project requirements for CSCI 4080 – Design for Educational Applications. The project was supervised by Dr. Michael Miljanovic; TA Cole Craven provided feedback at the proposal and prototype stages.

Correspondence concerning this article should be addressed to the Department of Computer Science, Ontario Tech University, 2000 Simcoe Street North, Oshawa, ON L1G 0C5.

Abstract

Signly is a Flutter-based mobile and web application built to help beginners learn American Sign Language (ASL). We created it because there are not many good digital tools for learning ASL, and the ones that exist tend to be repetitive and poorly designed. The app walks learners through four structured modules whose order adapts to the learner's experience level during sign-up, supports multiple interaction types including quizzes, scenario practice, error analysis, and camera-based signing, and includes accessibility settings grounded in Universal Design for Learning (UDL). It is aimed at total beginners, including students, family members of Deaf individuals, and professionals who want to communicate better in their workplace. The camera feature provides live feedback on signing attempts, automatically flagging incorrect responses into a dedicated Mistakes category so learners can revisit trouble signs in the Practice module. We also describe a planned evaluation using a pre-test and post-test design alongside heuristic and peer usability testing. Early feedback from peers was positive, particularly around the interface and the lesson flow. Future priorities include persistent settings, more video content, real user authentication, and a formal summative evaluation.

Keywords: American Sign Language, mobile learning, Universal Design for Learning, Bloom's Taxonomy, educational technology

Signly: An American Sign Language Learning Application

Sign language is a complete, natural human language used by Deaf and hard-of-hearing communities around the world. American Sign Language is one of the most commonly used sign languages in North America, with somewhere around 500,000 or more users in the United States and Canada (Mitchell et al., 2006). Even so, most popular language learning apps like Duolingo, Babbel, and Rosetta Stone focus entirely on spoken languages and leave ASL learners with very few good digital options.

The apps that do exist for ASL tend to have the same problems. They rely on repetitive multiple-choice questions, use static image flashcards instead of video, have outdated or plain interfaces, and offer no way to practice actually signing in real time. Lingvano is a good example of this. It gives you binary-choice questions that you can pretty much guess your way through, and the interface is noticeably bare with a lot of wasted space and not much to keep you engaged (Signly Project Team, 2026a). This gap in the market was the main motivation behind building Signly.

Signly is a Flutter-based app for Android, iOS, and web that offers guided lessons, a variety of interactive activity types, and a camera mode for live signing feedback. The lesson design follows Bloom's Revised Taxonomy (Anderson and Krathwohl, 2001) to make sure learners are being challenged at progressively deeper cognitive levels, not just memorizing and forgetting. The app is also built around Universal Design for Learning, which is a framework that encourages flexible instructional design to reduce barriers and support a wider range of learners (CAST, 2018).

This report walks through the background and motivation for the app, the full design of its features and interface, and a detailed plan for evaluating whether it actually works.

Learning Outcomes

Signly is organized around four core modules: Greetings and Introductions, Everyday Expressions, Family and Relationships, and School and Classroom. The order in which these modules are presented to the learner is not fixed. During sign-up, each user goes through a personalization flow where they select their experience level, learning purpose, and topics of interest. Based on those answers, the app surfaces the most relevant module first, meaning a learner focused on workplace communication will see a different starting point than someone learning to connect with a family member. A daily goal timer is also set during onboarding, and the app tracks progress against that target each session.

Across these modules, Signly targets five main learning outcomes. Learners should be able to recognize and correctly identify a working vocabulary of common ASL signs within their chosen topic areas after finishing the core modules. They should be able to produce simple signs using the camera practice feature and receive live feedback on their attempts. They should be able to select the right sign for a given real-world situation. They should be able to identify and reflect on signing errors through structured comparison tasks. Finally, they should develop a foundational understanding of ASL and Deaf culture as it comes up naturally in lesson content.

Target Audience

The main audience for Signly is people who are completely new to ASL and want to learn for personal reasons, people taking an ASL or Deaf Studies course, and family members or friends of Deaf or hard-of-hearing individuals. There is also a secondary audience of people in customer-facing jobs, like healthcare workers, teachers, or retail staff, who want to be able to communicate with Deaf customers or colleagues. The app assumes no prior knowledge of ASL, though users can choose their experience level during onboarding if they have some background already.

Background

The following subsections outline the educational and technical context behind Signly, covering how ASL is acquired, the frameworks that guided the design, and how the app compares to existing tools.

ASL and Sign Language Acquisition

ASL has its own phonology, morphology, and syntax that are completely separate from English (Mitchell et al., 2006). Learning to sign is not just about memorizing handshapes. It involves building spatial reasoning, visual attention, and motor memory for the specific movements that distinguish one sign from another.

Research on second language acquisition has consistently shown that learners retain vocabulary better when they get meaningful input, immediate feedback on mistakes, and spaced repetition over time (Krashen, 1982). Digital tools tend to work well for language learning when they include active recall, varied practice formats, and some degree of learner

control (Godwin-Jones, 2011). Mobile apps are a natural fit for ASL specifically because the camera and touchscreen give learners a way to practice producing signs and get live feedback, but most ASL apps have not really taken advantage of this.

Universal Design for Learning

UDL is a framework developed by CAST that pushes for more flexible instructional design so that more learners can succeed, regardless of their background or abilities (CAST, 2018). It is organized around three ideas: giving learners multiple reasons to engage with material, presenting information in multiple ways, and allowing learners to show what they know through multiple channels.

Signly is designed with all three of these in mind. Engagement is built in through daily goals, streak tracking, real-world scenarios, and five different practice modes that learners can choose between. Representation is handled through video demonstrations with replay and slow-motion, written breakdowns of each sign, adjustable text size, high contrast mode, and optional captions. Action and expression is supported through quizzes, camera recording, scenario selection, error analysis, structured reflection, dominant hand settings, and haptic feedback.

Bloom's Taxonomy and Lesson Design

Bloom's Revised Taxonomy describes a range of cognitive skills from basic recall at the bottom up to evaluation and creation at the top (Anderson and Krathwohl, 2001). Good instructional design moves learners through these levels rather than keeping them stuck at the bottom doing rote memorization.

Signly's lesson flow maps directly onto this. Learners start by watching a sign demonstration, which targets the Remember level. They then identify the correct sign from options, which is Understand. They apply the sign in a real-world scenario, analyze errors by comparing correct and incorrect signing side by side, produce the sign themselves using the camera, and finally reflect on what they found difficult and what they would do differently. By the end of a lesson, learners have worked at every level of the taxonomy at least once.

Related Work and Competitive Analysis

Lingvano is probably the closest direct competitor to Signly, but as mentioned, it uses very basic multiple-choice questions and does not give learners much to work with in terms of interface design or varied practice (Signly Project Team, 2026a). Duolingo has started a sign language course but it is clearly more about keeping users engaged with streaks and points than about actually teaching signing skills in depth (Duolingo, 2023). Academic research going back to the late 1990s has shown that camera-based feedback can meaningfully improve sign production accuracy when combined with immediate corrective output (Starner et al., 1998), but very few consumer apps have implemented anything like this.

Signly sets itself apart by combining a structured Bloom's-aligned lesson sequence, seven different Reusable Learning Objects that cover multiple cognitive levels, a full UDL accessibility panel, and a live camera feedback system for sign practice that does not require any subscription or proprietary hardware.

App Design

This section describes the system architecture, onboarding flow, module structure, lesson design, mistake tracking, accessibility features, and the two user scenarios that guided the navigation design.

System Architecture

Signly is built in Flutter and Dart, which means it runs on Android, iOS, and Chrome without needing separate codebases. The app has 29 screens total, organized into five areas: authentication and onboarding, main navigation, the lesson flow, practice modes, and settings and support pages. A Python FastAPI backend handles camera-based signing feedback. There is also a mock mode built in that keeps all the features working even when the backend is not running, which is useful for demos and testing.

The lesson content is separated from the screen logic through a shared data model, which makes it relatively easy to add new signs, units, or lesson variations without touching the screen code itself.

Onboarding

When a new user opens Signly for the first time, they go through five screens before reaching the main app. Figure 1 shows the splash screen, welcome screen, and login screen. There is a brief splash screen with the logo, then a welcome screen that explains what the app does and gives users the option to sign up or log in. The account setup screen collects an email, username, and password, with optional Google or Apple sign-in. The preferences screen lets users pick their sign language variant, experience level, learning purpose, daily practice goal, and starting level. The final onboarding screen shows a personalized summary of their choices and suggests what to do first.

Figure 1a

Splash screen with app logo and tagline

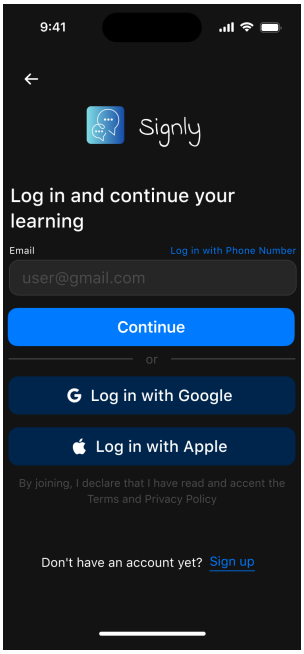

Figure 1b

Login screen for returning users

This approach was informed by UDL's Engagement principle, since letting learners set their own goals and preferences from the start helps with motivation and sustained effort. It also follows Nielsen Norman Group's guidance on user control and transparency (Nielsen, 1994).

Onboarding Preferences

Figure 2 shows the multi-step preferences flow. Users select their sign language region, then choose a level and set a daily learning goal. These choices directly shape which module is presented first and how the daily goal timer is calibrated for each session. A learner who sets a five-minute daily goal will see different pacing prompts than one who sets fifteen minutes.

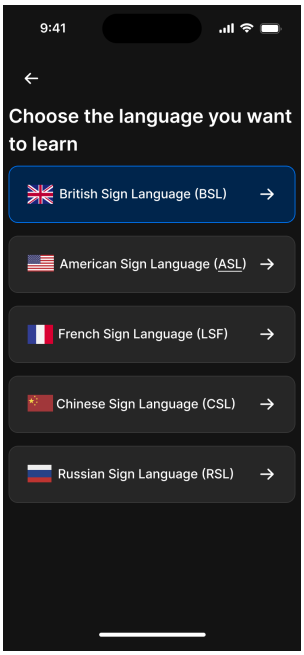

Figure 2a

Sign language variant selection (ASL, BSL, LSF, CSL, RSL)

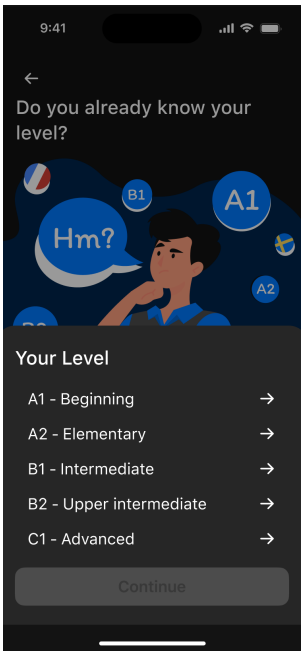

Figure 2b

Experience level and starting module selection

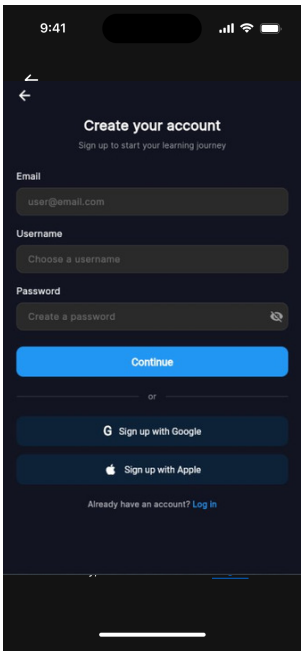

Figure 2c

Account creation with email and password

Module Structure

The four core modules in Signly are Greetings and Introductions, Everyday Expressions, Family and Relationships, and School and Classroom. As shown in Figure 3, each module card on the Learn tab displays the skill focus, lesson count, and current completion percentage. The order these appear for each learner depends on the experience level and topic preferences they selected during onboarding. A learner who indicated an interest in workplace communication, for example, may see Everyday Expressions surfaced before Greetings and Introductions.

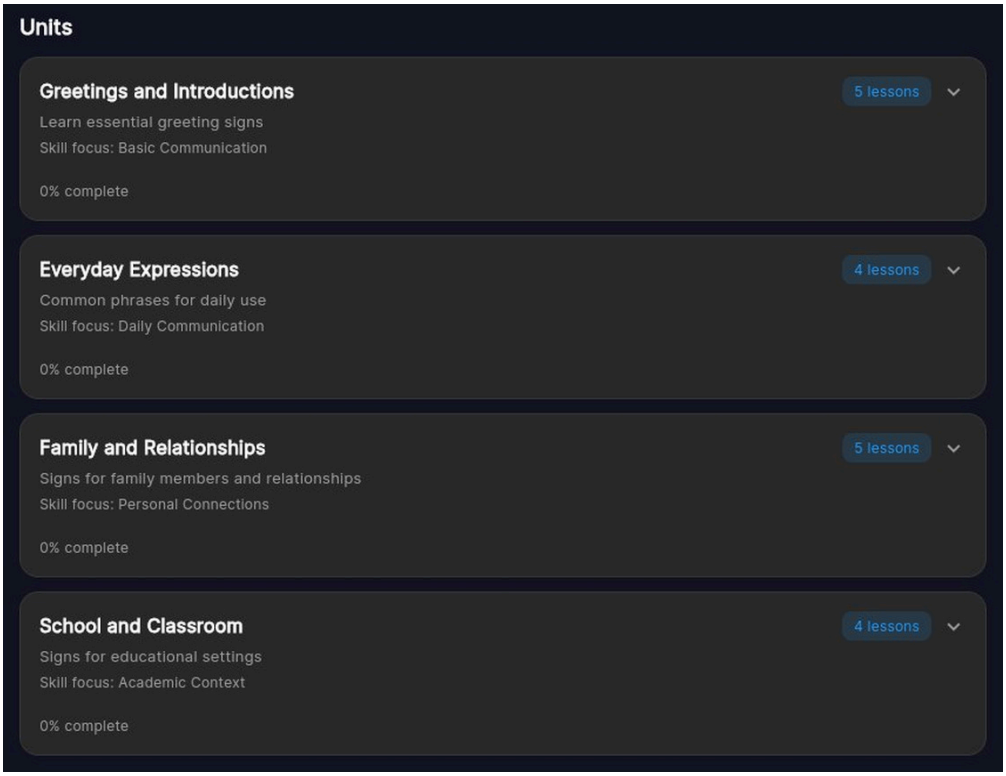

Figure 3. Module overview showing the four core units with skill focus and completion tracking.

Main Navigation and Lesson Flow

Once inside the app, users navigate through five tabs at the bottom of the screen: Learn, Practice, Vocabulary, Progress, and Profile. The Learn tab is where the structured lesson modules live. Figure 4 shows the Book Screen, which is the module overview page that learners land on after logging in. From there, tapping any lesson brings them into the eight-step lesson flow.

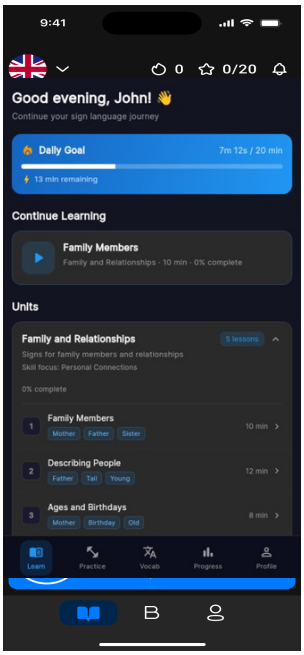

Figure 4a

Module overview screen (Book Screen) showing available units

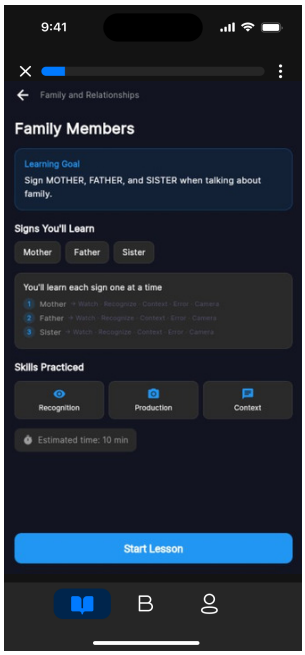

Figure 4b

Lesson intro with video player and Continue button

Each lesson follows an eight-step sequence that moves through Bloom's Taxonomy from bottom to top. A progress bar at the top of the screen shows where the learner is, and a confirmation dialog pops up if they try to leave before finishing. Table 1 shows how each step maps to a taxonomy level and what the learner is actually doing.

Table 1

Lesson Flow Mapping to Bloom's Revised Taxonomy

Step	Screen	Bloom's Level	Primary Activity
1	Lesson Intro	—	Learning objectives displayed using the ABCD model
2	Watch and Notice	Remember	Video demonstration with replay and 0.5x slow-motion
3	Recognize	Understand	Multiple-choice quiz with immediate corrective feedback
4	Context Practice	Apply	Select the right sign for a real-world scenario
5	Error Analysis	Analyze	Compare a correct and incorrect sign side by side
6	Camera Practice	Create	Produce the sign using the device camera with live feedback
7	Reflection	Evaluate	Self-assessment prompts about difficulty and improvement
8	Lesson Summary	—	Accuracy, completion status, and identified weak areas

Note. ABCD = Audience, Behavior, Condition, Degree. Every step requires active cognitive engagement.

Figure 5 shows examples of the interactive question screens within a lesson. The recognition and scenario screens show how learners are asked to work with signs in context, with immediate feedback shown at the bottom of the screen after each response.

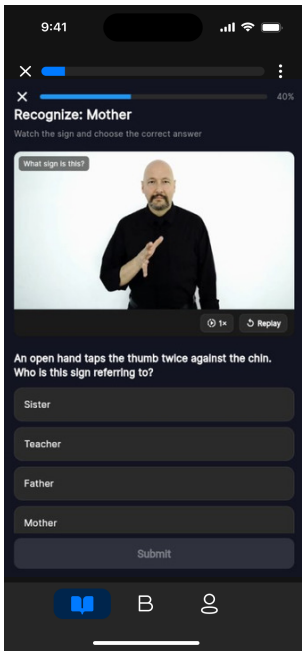

Figure 5a

Lesson recognition screen with sign image options

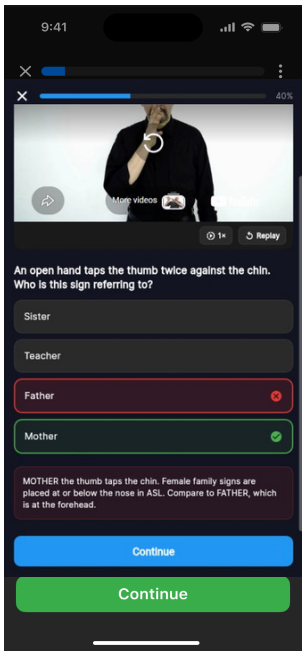

Figure 5b

Immediate correct/incorrect feedback after answering

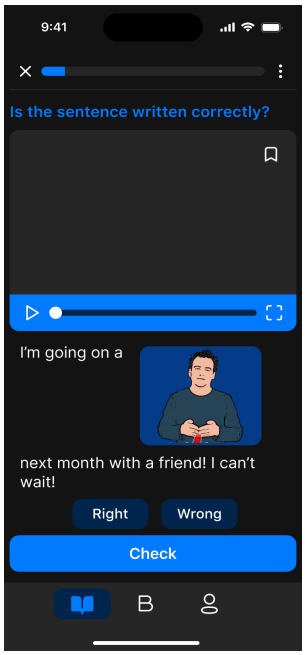

Figure 5c

Scenario-based context practice question

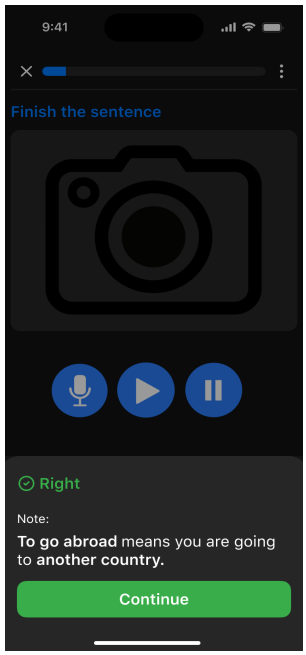

Figure 5d

Error analysis feedback with explanation

Mistake Tracking and the Practice Module

A key part of how Signly supports learning is what happens when a learner gets something wrong. Rather than simply moving on, the app detects incorrect responses and does three things automatically. First, it flags the sign as not yet learned, so it will not be counted toward lesson completion. Second, it adds the sign to a dedicated Mistakes category visible in the Vocabulary tab. Third, it notifies the learner that a mistake was made and shows them the correct answer.

This connects directly to the Practice module. Learners can open the Practice tab at any time and see a breakdown of the signs they have gotten wrong most often. The Mistake Review mode in the Practice tab pulls directly from this list, letting learners drill their trouble signs in a low-pressure environment outside of the main lesson flow. This design is grounded in spaced repetition principles, where revisiting difficult material at regular intervals is one of the most effective ways to move vocabulary into long-term memory (Krashen, 1982).

Reusable Learning Objects

The app has seven Reusable Learning Objects in total, which is more than the three or four that are typically expected. Each one is a standalone module with its own data model, so it can appear in both the lesson flow and the Practice tab without duplication. Table 2 lists them.

Table 2

Reusable Learning Objects in Signly

RLO	Type	Key Source File(s)
1	Interactive Video Demonstration	lesson_watch_screen.dart
2	Recognition Quiz	lesson_recognition_screen.dart; sign_interpretation_screen.dart
3	Scenario-Based Context Practice	lesson_context_screen.dart; scenario_practice_screen.dart
4	Error Analysis and Error Detective	lesson_error_screen.dart; error_detective_screen.dart
5	Camera-Based Production Practice	lesson_camera_screen.dart; camera_practice_screen.dart
6	Self-Reflection and Metacognition	lesson_reflection_screen.dart
7	Mistake Review and Spaced Repetition	mistake_review_screen.dart

Vocabulary and Progress Tracking

The Vocabulary tab gives learners a searchable list of all the signs they have encountered, grouped by category and sorted by performance. As shown in Figure 6, the screen surfaces three performance buckets at the top: Mistakes, Average, and Success, along with the learner's saved vocabulary list and popular sign sets. This gives learners a quick way to see where they need more work and to revisit specific signs without going back through a full lesson.

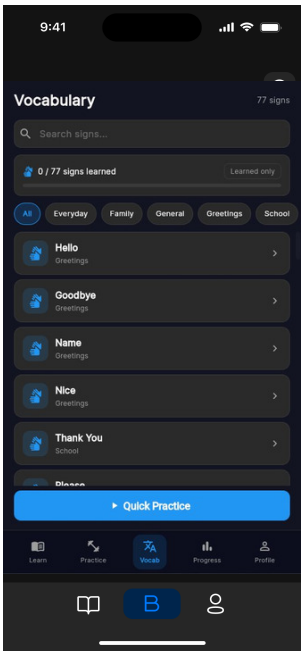

Figure 6a

Vocabulary tab showing performance breakdown and saved signs

Figure 6b

Profile tab with user stats and quick actions

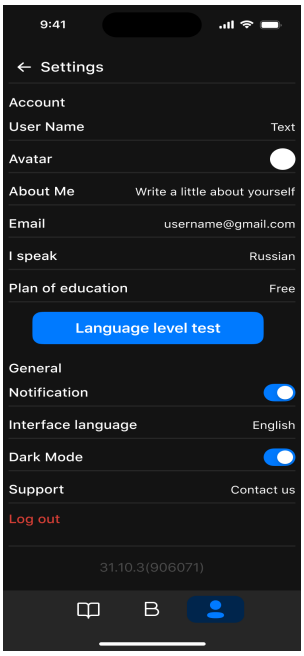

Figure 6c

Settings screen with account, general, and accessibility options

Accessibility and UDL Settings

The settings panel shown in Figure 6c covers all three UDL principles and is accessible from the Profile tab at any time, including during a lesson. Users can adjust text size across four levels, change animation speed, set video playback speed to 0.5x, 1x, or 1.5x, toggle high contrast mode, toggle captions, toggle haptic feedback, set dominant hand preference, and switch between dark and light mode. Full persistence of these settings across sessions is still in progress and is the top development priority for the next version of the app.

UI/UX Design

The app uses a dark theme throughout. The colour system uses blue for interactive elements and progress, green for correct answers and completions, red for errors, and orange for warnings. Spacing follows an 8-pixel grid with 20-pixel horizontal padding, and all cards and buttons have rounded corners at 12 to 16 pixels.

The interface is designed around Nielsen Norman Group's usability heuristics (Nielsen, 1994). Confirmation dialogs appear before any destructive action, including exiting a lesson, logging out, and resetting progress. Submit buttons are disabled until the user has made a selection to prevent accidental submissions. All primary action buttons are anchored to the bottom of the screen for consistent thumb-reach. Snackbars provide brief, non-intrusive feedback for completed actions.

User Scenarios and Storyboards

Two scenarios were used to validate the navigation structure during the design phase. The first scenario, shown in Figure 7, follows a new user through sign-up, onboarding, and their first lesson. The storyboard confirms that a learner can move from login through Module 1 and all three lessons before reaching a final quiz, and that they can exit back to the module page at any time. The second scenario, shown in Figure 8, follows an existing user logging in and finding the Free Mode camera feature by scrolling down from the module page. It confirms that a back button is always visible and that Free Mode does not require digging through menus. Both scenarios helped confirm that the structure is efficient and does not have unnecessary steps.

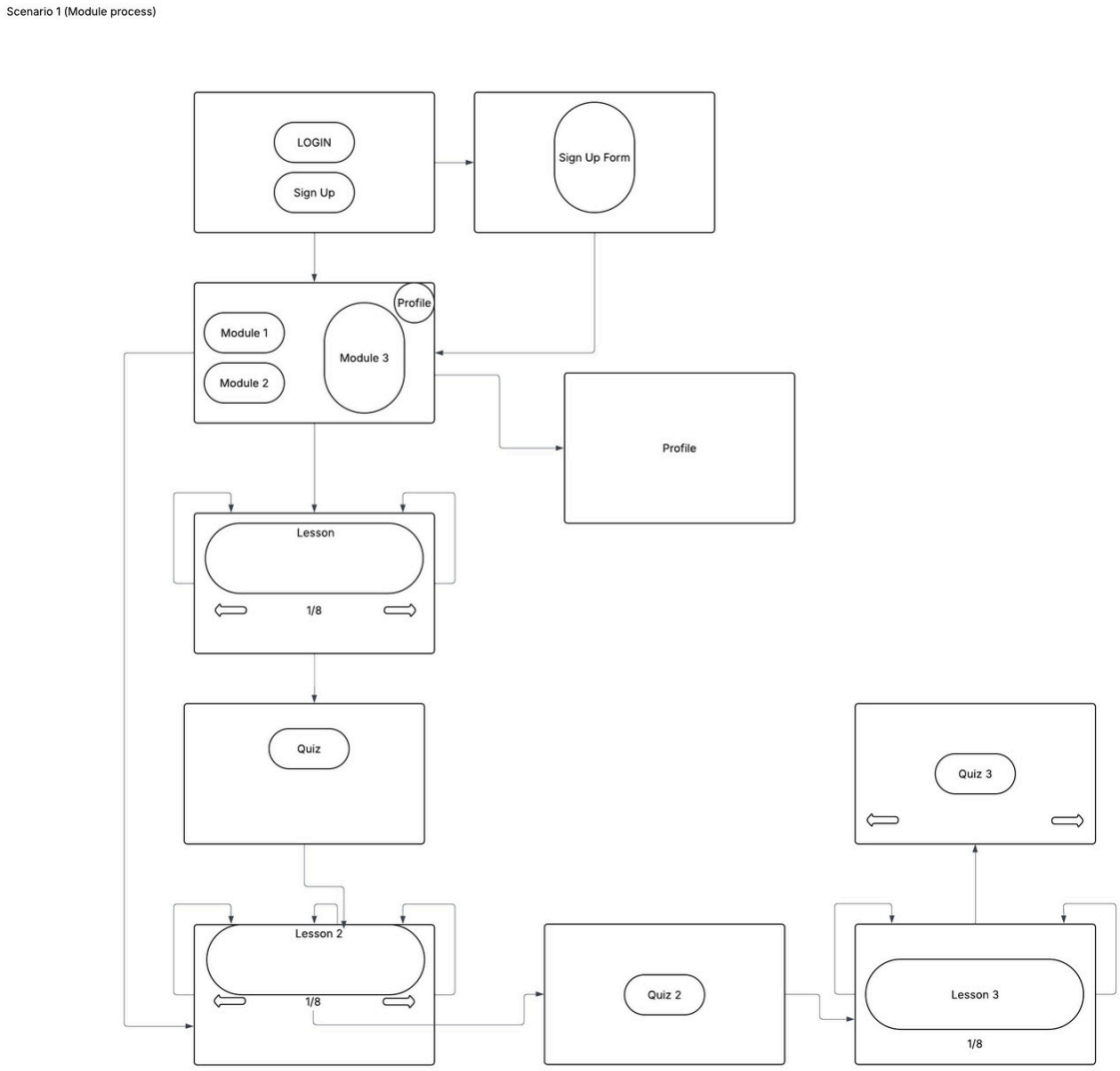

Figure 7. Scenario 1 storyboard: hierarchical flow showing login, module selection, lesson progression, and quiz completion.

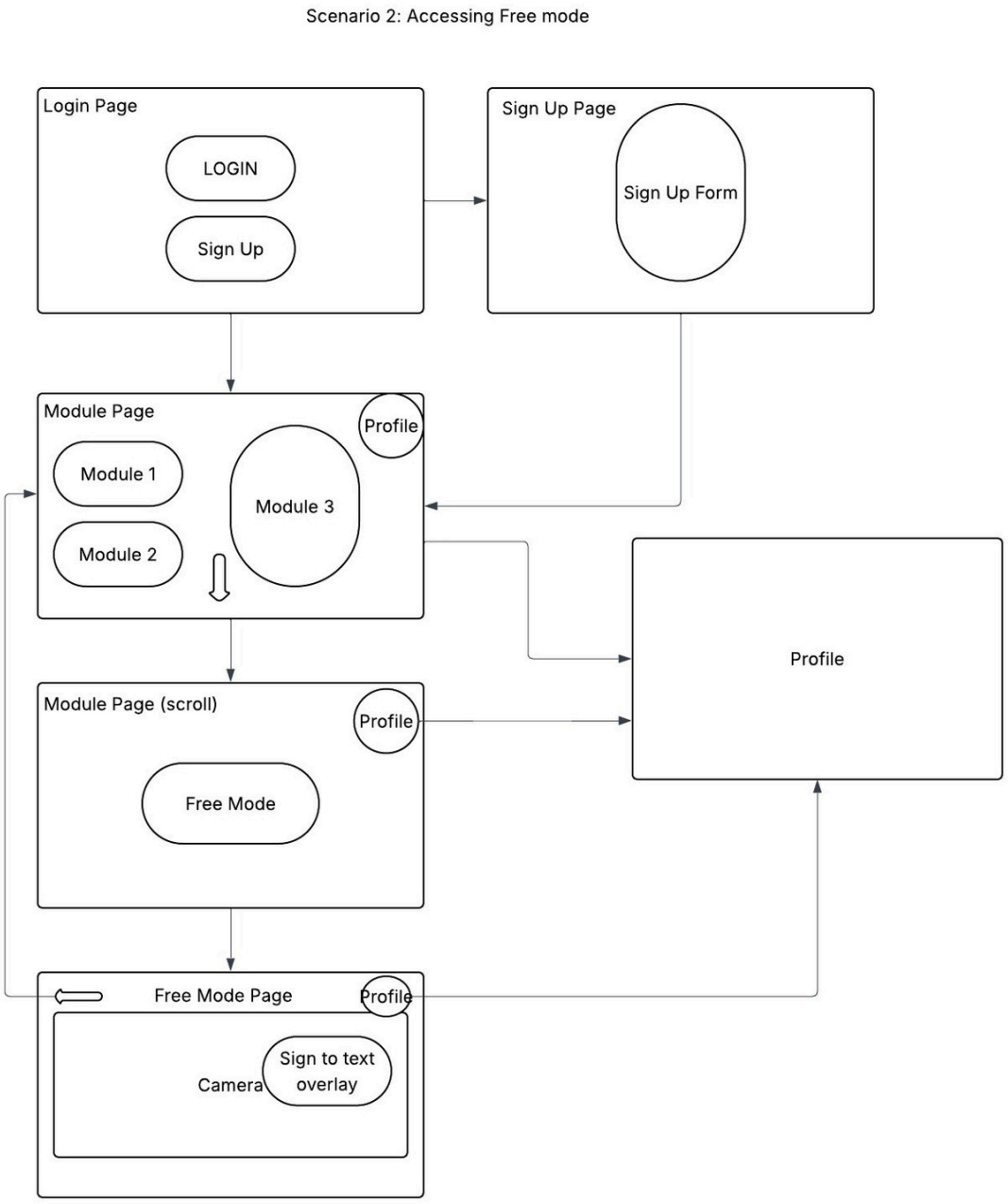

Figure 8. Scenario 2 storyboard: linear flow showing how a returning user accesses the Free Mode camera feature.

Evaluation

The evaluation plan addresses three questions: whether the app leads to measurable learning gains, whether the interface is usable, and whether the UDL accessibility features make a real difference for users with different needs.

Evaluation Goals

The evaluation plan is designed to answer three questions. First, does the app actually lead to measurable improvements in ASL recognition and production for beginners? Second, is the interface usable and consistent with good UX practices? Third, do the UDL accessibility features make a real difference for users with different needs?

Usability Evaluation

We did an initial heuristic evaluation using Nielsen and Molich's ten usability heuristics (Nielsen, 1994) as a checklist. The app did well on several fronts: confirmation dialogs are in place for all destructive actions, progress is always visible, navigation is consistent across screens, and feedback is immediate and clearly colour-coded. The two main issues we found were that settings do not yet persist across app restarts, and that video content beyond the first lesson uses placeholder assets instead of real sign demonstrations.

We also did informal peer testing with classmates. The general reaction was positive. People found the onboarding straightforward, liked the dark theme, and said the progress bar made the lesson feel manageable. A few people suggested adding a vocabulary shortcut on the Learn tab and making it clearer which lessons are locked.

Learning Outcomes Evaluation Plan

For a formal evaluation, we plan to use a pre-test and post-test design with at least 20 participants who have no prior ASL experience. Participants will complete a baseline recognition test before using the app and a parallel test after finishing the three core modules. The main thing we are measuring is how accurately they can identify signs. We will also look at production accuracy using an expert rubric applied to camera recordings, self-reported confidence on a five-point scale, and how long people spend in the app per session.

The app already collects embedded data that can support this evaluation. The Lesson Summary screen captures accuracy per step, and the Progress tab tracks streaks and identifies areas of strength and weakness. The Mistakes category in the Vocabulary tab also provides a natural record of which signs each learner found difficult, which can be used to compare performance before and after targeted practice. These built-in measures will complement the external pre and post tests.

UDL Accessibility Evaluation

For the accessibility piece, we plan to recruit from three specific groups: people with low vision, people who have motor differences that affect their dominant hand, and people who have limited experience with educational technology. Each participant will complete a 30-minute think-aloud session and then fill out the System Usability Scale (Brooke, 1996). A score above 68 on the SUS is generally considered above average. We will compare scores across the three groups to figure out where the biggest accessibility gaps are and prioritize fixes accordingly.

Camera Feedback Evaluation

The live camera feedback feature also needs its own evaluation. We plan to assess how useful learners find the feedback during camera practice, whether the live response feels timely and clear, and whether the automatic mistake detection correctly identifies genuinely incorrect attempts without penalizing minor variation in production. Participants will be asked to rate the camera feature specifically on a short post-session survey, and observation notes will capture any moments where the feedback appeared to confuse or discourage the learner. In the meantime, the mock mode in the current build ensures the app can be fully demonstrated and tested without needing the backend to be running.

Conclusion

Signly brings together a Bloom's-aligned eight-step lesson flow, four adaptive content modules, seven Reusable Learning Objects, a full UDL accessibility panel, and a live camera feedback system in a single cross-platform Flutter app. That combination is not something the existing ASL app landscape offers, and we think it represents a meaningful step forward in how sign language can be taught digitally.

Summary of Contributions

On the technical side, the main contributions are the decoupled lesson data model that makes content easy to extend, the RLO architecture that lets the same components serve both guided lessons and independent practice, and the mistake tracking system that connects incorrect responses directly to a targeted practice queue. On the pedagogical side, the contributions are the explicit Bloom's mapping across every lesson step, the adaptive module ordering based on learner preferences, the daily goal timer that keeps sessions focused, the UDL settings panel that covers engagement, representation, and expression, and the reflection step that introduces metacognitive awareness into what would otherwise be a straightforward skill drill.

Future Work

The highest priority for the next version is making settings and progress actually persist between sessions using SharedPreferences or Firebase, since right now everything resets when the app is closed. After that, the main items are real authentication through Firebase Auth, expanding sign demonstration videos beyond the first unit, improving the reliability of live camera feedback across varied lighting conditions and backgrounds, and implementing a more sophisticated spaced repetition algorithm in the Mistake Review mode. The formal pre-test and post-test evaluation described above is also something we want to run as soon as the video content and persistence issues are resolved.

Signly shows that it is possible to build something pedagogically thoughtful, technically capable, and genuinely accessible within the scope of a course project. We hope it serves as a foundation for a more complete tool that could make ASL more accessible to a wider range of learners.

References

Anderson, L. W., and Krathwohl, D. R. (Eds.). (2001). *A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives*. Longman.

Brooke, J. (1996). SUS: A quick and dirty usability scale. In P. Jordan, B. Thomas, B. Weerdmeester, and A. McClelland (Eds.), *Usability evaluation in industry* (pp. 189-194). Taylor and Francis.

CAST. (2018). *Universal Design for Learning guidelines version 2.2*. https://udlguidelines.cast.org

Duolingo. (2023). *Duolingo ASL course beta*. Duolingo Inc. https://www.duolingo.com

Godwin-Jones, R. (2011). Emerging technologies: Mobile apps for language learning. *Language Learning and Technology, 15*(2), 2-11.

Krashen, S. D. (1982). *Principles and practice in second language acquisition*. Pergamon Press.

Mitchell, R. E., Young, T. A., Bachleda, B., and Karchmer, M. A. (2006). How many people use ASL in the United States? Why estimates need updating. *Sign Language Studies, 6*(3), 306-335.

Nielsen, J. (1994). Enhancing the explanatory power of usability heuristics. In *Proceedings of the ACM CHI '94 Conference on Human Factors in Computing Systems* (pp. 152-158). ACM.

Signly Project Team. (2026a). *Signly – Project proposal* [Course project documentation, CSCI 4080]. Ontario Tech University.

Signly Project Team. (2026b). *Signly – App prototype report* [Course project documentation, CSCI 4080]. Ontario Tech University.

Starner, T., Weaver, J., and Pentland, A. (1998). Real-time American Sign Language recognition using desk and wearable computer-based video. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *20*(12), 1371-1375.